

PERFORMANCE ASSESSMENT OF DATA AND TIME-SENSITIVE WIRELESS DISTRIBUTED NETWORKED-CONTROL-SYSTEMS IN PRESENCE OF INFORMATION SECURITY

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Abstract- Distributed network-control-systems (D-NCS) are a network structure and components that are capable of integrating sensors, actuators, communication, and control algorithms to suit real-time applications. They have been gaining popularity due to their high potential in widespread applications and becoming realizable due to the rapid advancements in embedded systems, wireless communication and data transfer technologies. This paper addresses the issue of D-NCS information security as well its time-sensitive performance with respect to network security schemes. We use a wireless network based, robot navigation path tracking system called Intelligent Space (iSpace) as a D-NCS test bed in this paper. The paper classifies the delay data from every NCS module (sensors, actuators and controllers). We define performance parameters for this NCS test bed. Various system factors including network delay, system gain, affecting these performance parameters are recognized. Network security algorithms DES and 3DES are integrated with the application to secure the sensitive information flow. Standard statistical approach such as 2^k factorial experiment design, analysis of variance, hypothesis testing is used to study and estimate the effect of each factor on the system performance especially security features. Thorough experimental results, tables of detailed characterization and effect estimate analysis is presented followed by the discussion on the performance comparison of NCS with and without wireless security.

Index terms – Distributed networked-control-system, iSpace, data-sensitive wireless network, data security.

I. INTRODUCTION

A distributed networked-control-system (D-NCS) consists of distributed sensors, distributed actuators, and distributed computing processors over a communication network - Internet in this paper – to achieve almost limitless virtual space and making the system scalable. Research in tele-operation, initiated with the concern of safety in hazardous environments such as spaces and nuclear reactor plants, gave rise to a wide variety of applications like monitoring and operating manufacturing plants, robot navigations, space projects, nursing homes, military applications and many more over the years. The synergistic collaboration between industry and academia has led to advanced NCS especially in the field of automation. The implementation became possible and easy after the tremendous increase in the deployments of wireless systems. But the wireless medium is susceptible to easy intercepting, it is critical to protect transmitted data from unauthorized access and modifications in wireless systems. Wireless network security includes essential elements in Internet security devices that provide traffic filtering, integrity, confidentiality, and authentication. Therefore data sharing, and data/network security is of utmost concern in D-NCS considering the time and data sensitive applications. Most of these NCS that are not designed

with security protection can be vulnerable to network attacks nowadays, so there is a growing demand of efficient and scalable intrusion detection systems (IDS) in the network infrastructure of industrial plants. However, information security and data sensitivity issues have not been addressed enough to be applied in a time-sensitive D-NCS. Using security in an NCS gives rise to many topics like network architecture to support security for NCS, the performance assessment for NCS with security etc. Chi-Ho and Kwong [8] propose an efficient and biologically inspired learning model for anomaly intrusion detection in the multi-agent IDS of industrial plants. Creery and Byres [13] present methods to determine and reduce the vulnerability of NCS to malicious intrusions through thorough assessment of process control networks. Xu et al. [9] developed core architecture to address the collaborative control issues of distributed device networks under open and dynamic environments by adopting policy-based network security technologies. However, it is highly required to study the performance effects in NCS due to different security additions first before we can find out an optimum solution to have a time-sensitive as well as secure control over network. Gupta et al. [10] characterize the wireless NCS application on the basis of security effect on NCS performance to show the trade-off between security addition and real-time operation of D-NCS. To display the effect of addition of security in NCS, we developed a closed-loop network-controlled path tracking navigation system called intelligent space (iSpace) as a testbed. Path tracking error and total time to track the path are considered to be the performance defining parameters. iSpace consists of different modules such as image processing, automation control, and communication for navigation of a differential drive unmanned ground vehicle (UGV) in an indoor 2-D environment. We identify the system parameters affecting the system performance and then illustrate the procedure to characterize using 2^k factorial experiment design for time-sensitive applications with security considerations on iSpace platform. That is followed by the analysis of variance of the effect of each NCS parameter on the system performance. The statistical approach employed in this study is not restricted to the iSpace testbed at NCSU but applicable to any data- and time-sensitive wireless distributed network control system with consideration of information security.

II. SYSTEM DESCRIPTION AND IMPLEMENTATION

In the wireless systems, several security protocols such as wired equivalent privacy (WEP) and 802.1x port access control with extensible authentication protocol (EAP) support are proposed to address security issues[3][2]. Moreover, due to the strong security it provides in wired networks, the IP security protocol (IPsec) is considered as a good option for wireless systems as well. Some basic security features associated with these security protocols as defined by IETF for wireless systems are [6][1]: Authentication, Confidentiality, Integrity and non-

repudiation. In this paper, we are using one of the kinds of NCS - Intelligent Space (iSpace) - to characterize for confidentiality aspect of security service in D-NCS. We have used iSpace as a network based intelligent control platform for our experiments for two different security protocols so that we can characterize the system behavior.

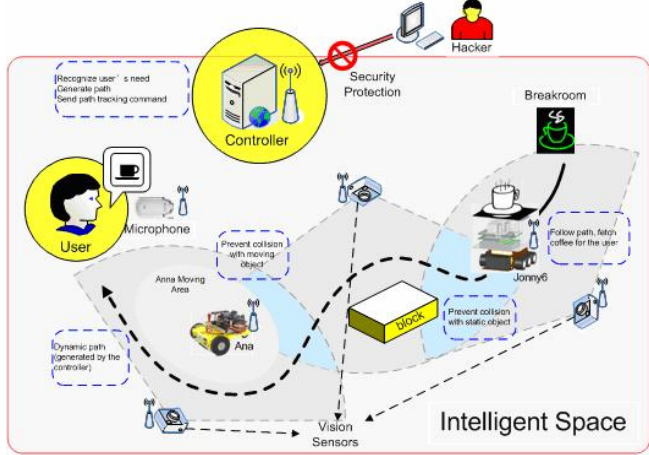


Figure 1: Describing iSpace concept as a D-NCS

A. iSpace as a D-NCS testbed

iSpace is a networked-controlled path tracking integrated navigation system for a UGV as shown in Figure 3. The destination point is chosen by the user via any remote computing interface in the world by accessing the iSpace GUI (Figure 2).

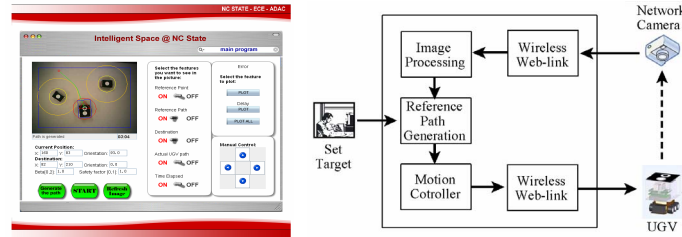


Figure 2: The graphical user interface for iSpace. Figure 3: iSpace - prototype of the NCS used as a test-bed.

(i) Overhead network cameras are used as global sensors. They collect the top-view image of the 2-D space of interest and send it to the main controller via internet for processing.

(ii) Differential drive UGVs (Figure 4) are the navigator in iSpace. Eq(1) defines the motion of a differential drive UGV. The UGVs receive commands from the main controller to initiate the movement to reach the destination specified by the user through the GUI. ω is the angular velocity of UGV, v is the linear velocity of UGV, and ω_l , ω_r are the angular velocities of left and right wheel respectively.

$$\begin{pmatrix} \omega_r \\ \omega_l \end{pmatrix} = \begin{pmatrix} 1/r & W/2r \\ 1/r & -W/2r \end{pmatrix} \begin{pmatrix} v \\ \omega \end{pmatrix} \quad \dots(1)$$

Figure 4: Model of differential drive UGV.



Figure 5: The UGV in iSpace - Betty

(iii) Main network controller (Linux Kernel 2.4.18-3 i686) has all the functional modules to operate the system. It is connected to the global sensors i.e. cameras and the actuators i.e. the UGV Betty in the space of interest through network. It also hosts the GUI.

(iv) UGV local controller: One of the major tasks of the local controller residing in the computing processor on the UGV is to receive mobility commands from the main controller via network and drive the actuators (servo motors) for the UGV.

Figure 5 depicts the whole control flow diagram for iSpace. The iSpace structure and the processing modules are explained in details in [4].

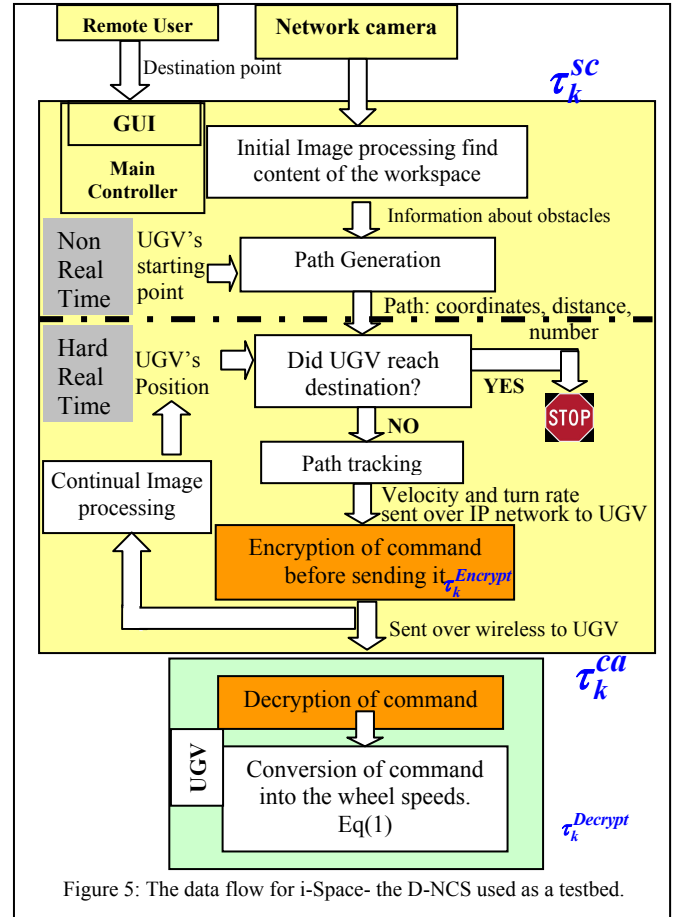


Figure 5: The data flow for iSpace- the D-NCS used as a testbed.

As the application can also be controlled over IP, we can use TCP/IP or UDP/IP for the communication between the sensors, actuators and the controller. TCP is slower than UDP because it also caters for the acknowledgement of the data received. In a time-sensitive application of iSpace, therefore, we send the sensor data with TCP protocol because the sensor data – image in case of iSpace – having all the needed information about the UGV workspace, will decide the course of processing and control commands generation by the main controller. However, the commands from the main controller to the actuator – the UGV in case of iSpace – are sent using the UDP protocol. The motion control commands are time sensitive. Once the interval time (Δt) is passed the old control command might have become obsolete and the new position of the UGV needs new motion commands to keep the UGV on path. It is one way

communication. If the packet is dropped, the UGV will receive the next command packet corresponding to the next updated position and act accordingly, all packets carrying the time sensitive information about the real-time motion of the UGV.

B. Data Encryption

Encryption algorithms for wireless systems can be broadly classified into two categories: symmetric and asymmetric ciphers. Encryption algorithms, which require a single cryptographic key to encrypt and decrypt the data, are termed as symmetric ciphers. The Data Encryption standard (DES) and the Advanced Encryption Standard (AES) are two symmetric ciphers which are considered secure for wireless systems [2][5]. However, the issue associated with symmetric ciphers is how to transmit cryptographic keys securely to the legitimate recipient. On the other hand, asymmetric ciphers are the ones which require different cryptographic keys for encryption and decryption of data. For example, public key encryptions, such as RSA, employ asymmetric ciphers. In public key encryption, each user has a pair of public and private keys. The public key is published while the private key is kept confidential. Messages are encrypted using the public key of the recipient, and can be decrypted only by the private key of the recipient. As private key is never known to others, this method eliminates the problem of transmitting secure keys to receivers. However, the critical issue associated with public key encryption is the high computational power required for encryption and decryption [1]. Thus public key encryption is used only for exchanging cryptographic keys securely in real time systems, whereas symmetric ciphers are used for actual data encryption and decryption.

In this work also, we implement DES and 3DES (a stronger variant of DES) for encrypting and decrypting the data securely over NCS. Data Encryption Standard (DES) incorporates a 64-bit key to encrypt and decrypt the transmitted data. However, as the least significant bit of each byte in the key is a parity bit, that bit from each byte is ignored during actual encryption and decryption process, leading to reduction of 8 bits from 64-bit key. Therefore, DES uses 56-bit key effectively [2]. Besides, DES, being a block cipher, operates on 64-bit blocks of plaintext as input and outputs 64-bit blocks of encrypted blocks. 3DES is very similar to DES and requires three 64-bit keys to provide stronger security than DES. In 3DES, first each 64-bit block is encrypted by using first 64-bit key, then decrypted using second 64-bit key and then encrypted again by using third 64-bit key. For decryption at the receiver side, the entire process is reversed to obtain the original form. Consequently, 3DES is 3 times slower than DES [5][6]. DES incorporates different modes of operations such as ECB (Electronic Code Book), CBC (Cipher Block Chaining), CFB (Cipher Feedback) and OFB (Output Feedback). As ECB is considered the fastest mode of operation, it is used commonly in real time systems [6]. In this work, we use DES and 3DES with ECB mode to reduce computational processing.

III. PROBLEM DEFINITION AND DESIGN OF EXPERIMENT

Stability and reliability are two of the main concerns for a D-NCS. When a UGV is controlled over a network, its performance can be degraded by network-induced delays and

other network factors, and the system can even become unstable. Similarly, because of the lack of security, the data can be maliciously used affecting the reliability of the operation. These concerns considered simultaneously will have different effects on the system described. And these concerns can be addressed only when we characterize and estimate different network parameters' effects on the performance of the remote controlled, network based time sensitive application.

A. Delay Classification of the System

First, we illustrate the different delays which have their effects on the functionality of the entire system. On a broader level, the delay classification can be done as Transportation / Network delay component and Computational / Processing delay component. Their definitions and significance for a distributed NCS is explained as below.

Transportation / Network delay component (τ_k^{ND})	
Sensor to controller delay (τ_k^{sc})	$\tau_k^{sc} = \tau_k^{imageacq} + n \cdot m (\tau_k^{RTT})$ $\tau_k^{imageacq}$ - The time between the main controller requests the camera for the image τ_k^{RTT} - Round trip network time. m - Number of packets needed to send the complete image information. n - Number of cameras to access to get the entire workspace image.
Controller to actuator delay (τ_k^{ca})	$\tau_k^{ca} = \tau_k^{RTT} / 2$
Computational / Processing delay component (τ_k^{CD})	
Non-real time computational delay (τ_k^{NRc})	$\tau_k^{NRc} = \tau_k^{imeproc} + \tau_k^{pathgen} + \tau_k^{sysproc}$ $\tau_k^{imeproc}$ - Initial image processing time. $\tau_k^{pathgen}$ - path generation time. $\tau_k^{sysproc}$ - system processing time.
Real time Computational delay (τ_k^{Rc})	$\tau_k^{Rc} = l \cdot (\tau_k^{imeproc-r} + \tau_k^{pathtrack} + \tau_k^{sysproc-r})$ l - Number of times the control loop runs before the UGV reaches the destination.
Confidentiality delay	
$\tau_k^{Encrypt}$	Encryption time of the command on the controller side.
$\tau_k^{Decrypt}$	Decryption time of the command on actuator side.

B. Performance parameters

In this paper, the application we are considering is networked controlled path tracking. The requirements are that the UGV should track the reference path as closely as possible.

- (1) The distance between the UGV's position and the reference path generated by the main controller is calculated at each Δt , which constitutes the distance/tracking error.
- (2) The next condition is the time bound. The UGV has to reach the destination as quickly as possible. Therefore following two performance measure parameters are defined.

y_1 : e_{track} - Average tracking error from the generated path,

y_2 : T_{total} - Total time to reach the destination / complete tracking.

C. Experiment setup

Before identifying the factors which will affect the system performance, we present the typical delay data in the iSpace system. τ_k^{sc} and τ_k^{rc} in each loop is observed for around 40 experiment runs.

Mean (τ_k^{sc}) = 0.2185 sec, Variance (τ_k^{sc}) = 2.5 e-4.

Mean (τ_k^{rc}) = 0.0223 sec. Variance(τ_k^{rc}) = 1.5 e-4.(2)

Thus τ_k^{rc} time constitutes of only 9 to 10% of the total average system loop time because the data size from controller to actuator is 12 bytes in size comparing to the 6 kb data (image) size from sensor to controller. The purpose of the paper is to analyze the effect of security feature on the NCS performance if any and signify it. Therefore we added the confidentiality for the control data from controller to actuator. Thus confidentiality is added to the part of the system which contributes to only 10% of the system operation and delay overhead. Thus the effect due to the security feature added to the very small part of the system can also be characterized and studied. Figure 6 presents the comparison of entire control loop time delay with and without security addition.

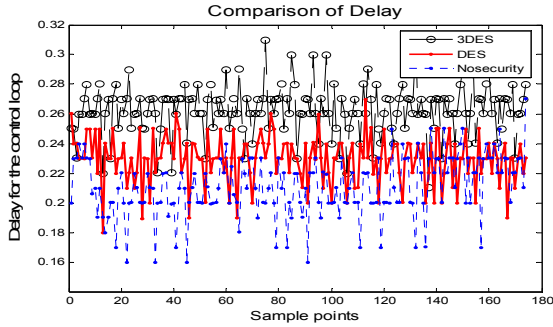


Figure 6: Control loop delay plotted for one experiment run

The following factors will affect the iSpace system performance:

NCS parameters

- (i) Network delay (N_{delay}): N_{delay} is random in nature and can be represented by stochastic models.
- (ii) Processing delay (P_{delay}): P_{delay} depends upon the algorithm complexity.
- (iii) System gain (β): Gain Scheduling Middleware gain [11]
- (iv) Security in loop (U_{sec})

The N_{delay} factor on Internet is random in nature. Therefore to repeat the experimental environment and to have control over these factors, we setup the system as shown in Figure 7. Instead of wireless communication, the sensors to controller and the controller to the UGV are connected using the hardwired connection. We connected the actuator to the controller with a delay generator in between so that we can fully control the delay in the experiment. The delay generator is PC-based implementation to simulate network delay. The delay generator receives the packets from the controller in a buffer format and it sends the packets to the UGV on the serial channel after applying a certain delay which is user defined. Considering negligible delay and no packet drops on the serial port, we can simulate the network delay τ_k^{ca} by giving different inputs of Δt to the network delay generator.

From Eq(2), since the variances are small and insignificant in comparison with the mean, the sensor to controller delay as well as processing delay are considered to be constant for current operation of iSpace. The experimental system is also designed such that the system gain β is taken as a constant user input while running the experiment. Thus we are running the NCS experiment in a controlled environment where each significant factor affecting the system can be varied, to study its effect on the system performance repeatedly.

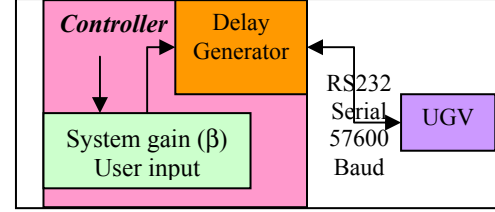


Figure 7: Modified experimental setup to control the experiment factors

Application dependent factors

In this paper, path tracking is the NCS application we are dealing with. Quadratic curve fitting path tracking controller is implemented as the path tracking controller [12]. The basic principle of this control algorithm is that a reference point is moved along a desired path so that the length between the reference point and the UGV is kept in some distance (d_0). Control (velocity) commands - speed (v , in cm/s) and turn rate (ω , in rad/s) - are generated for the UGV to reach that reference position from the current position. d_0 is calculated by Eq(3). Therefore the factors affecting e_{track} and T_{total} the most are (i) Path curvature (κ) and (ii) Maximum look-ahead distance for the path tracking (S_{max}).

$$d_0 = \frac{S_{max}}{1 + \beta|\kappa|} \quad (3)$$

As S_{max} and κ are path tracking parameters [4], they are application dependent factors, which might be different in case of different NCS application. But all other factors are common for all NCS applications. Therefore to study the effect of network delay, security features in any NCS, we fix the values for the application dependent factors i.e. S_{max} and κ for the current system operation. We define a fixed path with predefined curvatures for all experiments as shown in Figure 8. The path consists of pieces of different path curvatures.

$P = P1 \cup P2 \cup P3 \cup P4 \cup P5$, where, $\kappa(P1) = 0$, $\kappa(P2) = 1 \text{ m}^{-1}$, $\kappa(P3) = 2 \text{ m}^{-1}$, $\kappa(P4) = 4 \text{ m}^{-1}$, $\kappa(P5) = 0$

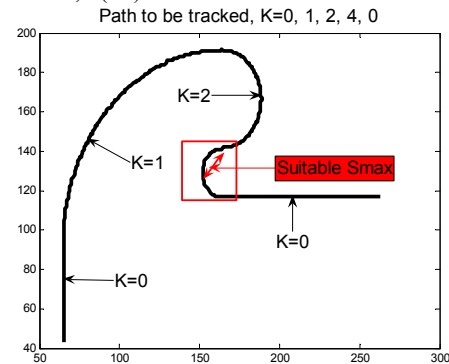


Figure 8: Path P to be tracked with curvature pieces as $\{0, 2, 4, 8, 0\} \text{ m}^{-1}$

To find a suitable value of $S_{\max}$ for the current operation of iSpace, we first define different levels of the $S_{\max}$ for path tracking operation in iSpace. To decide the range of $S_{\max}$ we intuitively decide the lower and higher value suitable for $S_{\max}$. The look-ahead distance should be ideally chosen such that the interpolated path between the current and the reference point matches the actual path as close as possible. In the predefined path P, $\kappa_{\max} = 4 \text{ m}^{-1}$ as $r_{\max} = 0.25 \text{ m}$. We have half circle with $\kappa_{\max}$ in P. For practical purposes $S_{\max}$ should not be greater than quarter the perimeter i.e. $0.5\pi r_{\max}$ (as shown in Figure 8). Therefore,

$$\max(S_{\max}) = 0.125\pi = 0.3927 \approx 0.4 \text{ m}.$$

Therefore we vary $S_{\max}$ through 5 levels $\{0.05, 0.1, 0.2, 0.3, 0.4\}$. The path tracking experiment is run repeatedly 5 times with each value of $S_{\max}$ with other factors kept constant at the values $N_{\text{delay}} = 0 \text{ sec}$, $\beta = 1$, $U_{\text{sec}} = \text{No security}$. After analyzing the e_{track} and T_{total} we decide that $S_{\max} = 0.2 \text{ m}$ is a suitable value for tracking of the path P (Figure 9).

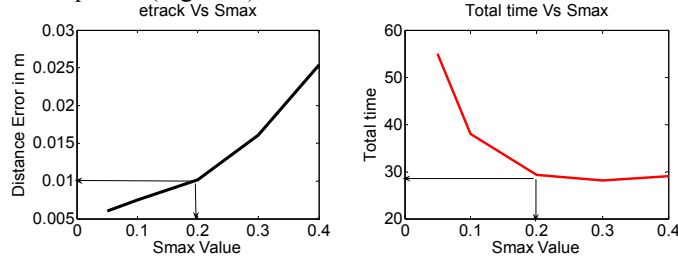


Figure 9: Mean of T_{total} and e_{track} Vs $S_{\max}$ over 5 experiment runs. $S_{\max} = 0.2 \text{ m}$ is suitable looking at the lowest T_{total} and comparatively low e_{track}

D. 2^k factorial design

After choosing suitable value for application dependent factors and setting up the experiment so that other NCS factors are controllable, we use statistical experiment design approach to estimate the effect of factors on the NCS performance. We have $k=3$ factors which are controllable. Thus, we use the 2^k factorial design to estimate the effect of these factors' values on NCS performance measure [7]. For 2^k factorial design, we have to define low and high level for all the factors [7]. That is done as follows:

(1) N_{delay} (**Factor A**): Typical loop network delay observed when the iSpace is operated over wireless channel varies from 0.2 to 0.3 seconds. Generally speaking, the network delay is in the order of a few hundreds of millisecond on the Internet. Typical delays observed in US continent are 0.1 s to 0.6 s. Therefore we define lower limit of $N_{\text{delay}} = 20 \text{ ms}$ (negligible delay) and higher limit of $N_{\text{delay}} = 0.6 \text{ s} = 600 \text{ ms}$.

(2) System gain (β) (**Factor B**): $\beta = 1$ is considered to be the standard or typical value for the system gain. We take lower and higher limit for β to be 0.5 and 1.5 respectively. (Symmetric about $\beta=1$).

(3) Security in the loop (U_{sec}) (**Factor C**): As mentioned before, we take into consideration 3 types of security features. (i) No security (ii) DES encryption (iii) 3DES encryption. Therefore we have two sets of factorial design experiments.

Exp#1: No security in comparison with DES and

Exp#2: No security in comparison with 3DES.

The experiments are replicated 5 times ($n=5$). For detailed explanation of 2^k factorial design, please refer to [7].

Table 1: Low and High levels for each factor in the experiment

Factors	Factor Levels	
	Low(-)	High (+)
A: N_{delay}	20 ms	600 ms
B: System gain (β)	0.5	1.5
C: U_{sec}	NoSecurity	Exp#1 – DES Exp#2 – 3DES

In this section, we estimate the effect on e_{track} and T_{total} due to the three factors separately as well as the interaction between the factors if any. As $k = 3$, we will run the experiment for $2^3 = 8$ treatments. As given in [7], the sum of n replicates for these 8 treatments are represented as $((1), a, b, ab, c, ac, bc, abc)$. To explain in brief what each treatment name stands for - (1) is the treatment with all factors at their low levels i.e. $A = 20 \text{ ms}$, $B = 0.5$ and $C = \text{No security}$. a stands for the treatment with factor A at high ($A = 600 \text{ ms}$) and factors B and C low level ($B = 0.5$ and $C = \text{No security}$) and so on.

Table 2: Total of all 8 treatments for e_{track} and T_{total} .

For $n=5$	Exp#1 e_{track} (m)	Exp#2 e_{track} (m)	Exp#1 T_{total} (sec)	Exp#2 T_{total} (sec)
(1)	0.0165	0.0165	109.3901	109.3901
a	0.0243	0.0243	193.3400	193.3400
b	0.0160	0.0160	60.4598	60.4598
ab	0.0296	0.0296	118.6400	118.6400
c	0.0247	0.0153	119.0548	112.9284
ac	0.0245	0.0100	175.8941	181.7400
bc	0.0146	0.0216	56.1463	56.7712
abc	0.1096	0.0992	119.0500	111.7600

Using the totals under the treatment combination shown in Table 2, we estimate the factor effects using standard formula for 3 factor 2^k factorial experiment as given in [7]:

Table 3: Main effect estimate on system performance with N_{delay} , System gain and Security addition.

Factor effects	Exp#1 T_{total}	Exp#2 T_{total}	Exp#1 e_{track}	Exp#2 e_{track}
A	30.64192	30.08097	0.008425	0.00793
B	-28.7346	-31.1092	0.004355	0.00477
C	-1.7407	-3.01476	0.003105	0.00172
AB	-3.49786	-4.82861	0.004805	0.00459
AC	-1.90092	-2.46184	0.002395	0.0019
BC	7.04662	4.67198	0.001825	0.00224
ABC	5.6436	4.31282	0.002315	0.0021

The magnitude of the factor effects in Table 3 indicates that the largest effect on T_{total} is due to the N_{delay} and System gain (β) followed by the interaction between β and U_{sec} . This is consistent for both DES and 3DES except that in case of 3DES, interaction between N_{delay} and β affects with the same degree. If we consider the e_{track} then, for DES, the largest effect is caused by N_{delay} , followed by the interaction between β and U_{sec} . However for 3DES feature, N_{delay} , β and interaction between them are the highest affecting factors. We observe that, effect of security addition on tracking error (Factor C) is almost 60 to 75 % of the effect due to other significant factors N_{delay} and β . The effect of security features on T_{total} is far less, approximately 4 to 13 % of the effect due to N_{delay} and T_{total} .

E. Residual Analysis and Model checking

For each experiment, we recognize the most significant factors from Table 3 and try to fit linear regression models given in Eq(4). We plot the residual normal probability plot for model checking.

$$\begin{aligned}\hat{y}_{DES_etrack} &= \beta_0 + \beta_1(A) + \beta_2(B) + \beta_{12}(A)(B) + \beta_3(C) \\ \hat{y}_{3DES_etrack} &= \beta_0 + \beta_1(A) + \beta_2(B) + \beta_{12}(A)(B) + \beta_{23}(B)(C) \\ \hat{y}_{DES_Total} &= \beta_0 + \beta_1(A) + \beta_2(B) + \beta_{23}(B)(C) + \beta_{123}(A)(B)(C) \\ \hat{y}_{3DES_Total} &= \beta_0 + \beta_1(A) + \beta_2(B) + \beta_{12}(A)(B) + \beta_{23}(B)(C) + \beta_{123}(A)(B)(C)\end{aligned}\quad ..(4)$$

Based on Figure 10, we can assume the system to be a random effect linear model.

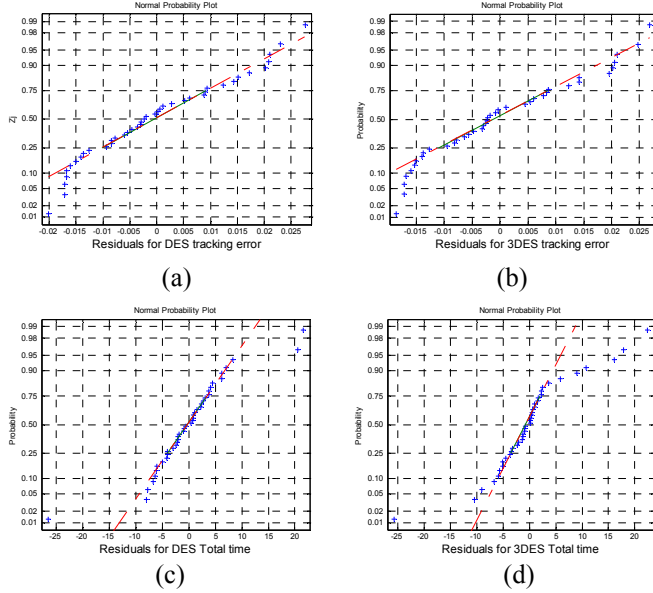


Figure 10: Normal probability plots of residuals from all 4 experiments. (a) DES_etrack (b) 3DES_etrack (c) DES_Time (d) 3DES_time

We therefore define the null hypothesis considering that designing of any NCS can be done without considering the effect of security additions on the system performance.

Thus H_0 : Security feature addition and its interaction with other factors do not have significant main effect on NCS performance. i.e. $\text{Effect}(C) \pm s.e.(\text{Effect})_{95\%}$ includes zero. Or $\beta_3 = \beta_{23} = \beta_{13} = \beta_{123} = 0$ i.e. the coefficients of the security features and its interaction with other factors in the linear model are zero.

H_1 : Security feature addition does affect the NCS performance or at least one of the $\beta_3, \beta_{23}, \beta_{13}$, and $\beta_{123} \neq 0$.

IV. HYPOTHESIS TESTING

The first step of the problems that we are trying to solve is to estimate the effect and analyze it to know the significance of the security feature effect on NCS performance as well as reason the findings. We will use two methods to determine which effects are non-zero. In the first method, we will calculate the standard errors of the effect and compare the magnitudes of the effects to their standard errors. Second method is Analysis of variance.

A. Standard error of effects

Let S_i^2 be the variance estimate of each treatment, $\bar{y}_i$ be the sample mean of the n observations in each treatment. $s.e.(\text{Effect})$ be the standard error of each estimated effect. (Eq 5).

$$\bar{y}_i = \sum_{j=1}^n y_{ij} / n \quad S_i^2 = \frac{1}{(n-1)} \sum_{j=1}^n (y_{ij} - \bar{y}_i)^2 \quad s.e.(\text{Effect}) = \sqrt{1/n^{2k-2} (S^2)} \quad \dots(5)$$

Table 4: Standard Error of effect of estimate for each experiment

Treatments	s.e. DES_etrack	s.e. 3DES_etrack	s.e. DES_time	s.e. 3DES_time
C	0.0008751	0.0006175	1.73788	1.57499
AC	0.0006556	0.0007111	2.10369	2.83575
BC	0.0005828	0.0002877	0.89413	0.44485
ABC	0.0015533	0.0015602	0.69225	1.7310

Testing of Hypothesis:

Looking at the $s.e.(\text{Effect})$, we observe that the treatment interval for only AC for total time and (-1.9009 ± 2.1037) and (-2.46184 ± 2.83575) include zero. These intervals are approximately 95% confidence intervals. All the factors individually and their interaction with other factors have effect on system performance. $\therefore H_0$ is rejected as estimate of effect of security addition and interaction of security addition with other significant factors such as N_{delay} and β with 2 standard deviation limits does not include zero. That means security feature addition does have effect on NCS performance.

B. Analysis of variance for Factor C - Security

Table 5: Analysis of Variance for effect of DES on e_{track}

Source of Variation	Sum of Squares	Degree of Freedom	Mean Square	F_0
C	9.64E-05	1	9.64E-05	2.4036
AC	5.74E-05	1	5.74E-05	1.4300
BC	3.33E-05	1	3.33E-05	0.8303
ABC	5.36E-05	1	5.36E-05	1.3361
Error	0.000320884	8	4.011E-05	
Total	0.0016919	15		

Table 6: Analysis of Variance for effect of 3DES on e_{track}

Source of Variation	Sum of Squares	Degree of Freedom	Mean Square	F_0
C	0.0000296	1	0.000029584	0.791
AC	0.0000361	1	0.0000361	0.965
BC	0.0000502	1	0.000050176	1.342
ABC	0.0000441	1	0.0000441	1.179
Error	0.0002991	8	3.739E-05	
Total	0.0015261	15		

Table 7: Analysis of Variance for effect of DES on T_{total}

Source of Variation	Sum of Squares	Degree of Freedom	Mean Square	F_0
C	30.30036	1	30.2024	0.114
AC	36.134968	1	44.2554	0.137
BC	496.54853	1	7.9948	1.882
ABC	318.50220	1	4.7922	1.207
Error	2110.8217	8	263.85271	
Total	20760.691	15		

Table 8: Analysis of Variance for effect of 3DES on T_{total}

Source of Variation	Sum of Squares	Degree of Freedom	Mean Square	F_0
C	90.8878	1	90.8878	0.317
AC	60.6066	1	60.6066	0.211
BC	218.2740	1	218.2740	0.761
ABC	186.0042	1	186.0042	0.649
Error	2294.2813	8	286.78517	
Total	21809.673	15		

Testing of Hypothesis:

We will reject hypothesis H_0 if $F_0 > F_{\alpha, a-1, N-a}$, $\alpha=1$ and $N-a=8$. $F_0(\text{DES_etrack_C}) > F_{0.25,1,8} \therefore H_0: \beta_3 = 0$ is rejected for DES tracking error case. $F_0(\text{DES_T}_{total_BC}) > F_{0.25,1,8} \therefore H_0: \beta_{23} = 0$ is rejected for DES total time case. However in other cases the effect of security addition is much less even considering 75% confidence interval as $F_0 > F_{0.25,1,8}$. The security addition, thus affect the system performance. The effect is comparatively less than the other factors, especially N_{delay} because of the following reason. We explained in the section III that security feature is added to only 10% of the system delay overhead.

$$\text{Mean}(\tau_k^{\text{Encrypt}})_{\text{DES}} \approx \text{Mean}(\tau_k^{\text{Decrypt}})_{\text{DES}} = 0.020, \text{Mean}(\tau_k^{\text{Encrypt}})_{3\text{DES}} \approx \text{Mean}(\tau_k^{\text{Decrypt}})_{3\text{DES}} = 0.055.$$

Thus the DES and 3DES encryption delay overhead are much less (9 to 18%) comparing to the operational and network delay of the system under observation (Figure 6). Therefore the effects due to U_{sec} is much less than the effect due to N_{delay} and β . However, we observed that security addition in only one part of the system operation which is only 10% part of overall the system performance also creates significant effect on system performance.

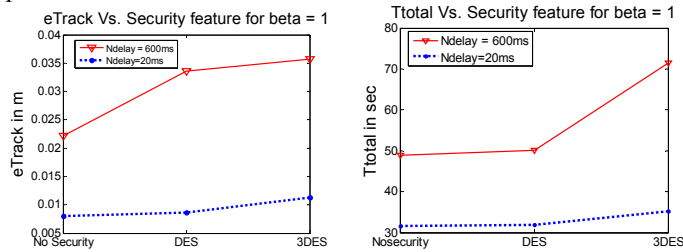


Figure 11: Comparison of performance parameters with respect to security addition and network delay for $\beta = 1$.

V. CONCLUSION AND FUTURE WORK

Through the experiments on the iSpace platform, which demonstrates a D-NCS, we characterized a data-sensitive wireless distributed NCS, network delay effect and the requirement for the network security. This paper puts forth a point that statically adding security algorithms in a time sensitive NCS would degrade the performance of the system to some extent, which is critical considering the stability of the NCS. Using statistical characterization approach we also observed that interaction between security features, network delay and system gain is an important factor towards NCS performance. We observed that security addition in only 10% part of overall the system delay overhead also creates significant effect on system performance. This trade-off between the security addition and

real-time performance of the system needs to be solved. As the security addition effect is much less than the effect due to N_{delay} we can say that security addition in NCS is very much possible and of utmost important. The security addition and the gain of the system, together also show significant effect on system performance. This can be further used to optimize the system performance by changing the system gain dynamically to compensate for security addition performance degradation. Thus, future work will involve investigating an optimum solution and between the reliability and offered by network security, stability and time and data-sensitivity required for any form of NCS. Therefore the characterization of any data-sensitive wireless NCS over different parameters is a key.

ACKNOWLEDGEMENT

The authors acknowledge the National Science Foundation (NSF) for sponsoring this research (Grant No. IIS-0426852). The opinions expressed are those of the authors and do not necessarily reflect the views of the NSF.

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